

NEWS RELEASE

TANDY CORPORATION/RADIO SHACK

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SUBJECT:

TRS-80® Model 16

FOR January 19 RELEASE

Radio Shack announces TRS-80® Model 16 Computer - uses 16-bit
MC68000 CPU & 8-bit Z-80 CPU - Model II compatible - advanced performance.

Radio Shack, a division of Tandy Corporation, has previewed the company's fastest, most powerful TRS-80 computer ever, featuring sixteen bit technology, dual processor architecture and a multi-user operating system. The new TRS-80® Model 16 is capable of 512K internal RAM memory storage and 2½ Megabytes of disk memory using two built-in all-new "thin line" double-sided 8-inch disk drives; additional external disk storage may also be added. This combination of large RAM and disk memory capacity will allow the Model 16 to use more sophisticated, user-friendly software with large file capacities.

The Model 16 desktop computer will be available second quarter 1982 in two versions, both with 128K memory. A single drive version with 1½ Megabytes of disk storage (26-6001) will be available for \$4999* at Radio Shack Computer Centers or the expanded computer departments of selected Radio Shack stores and dealers. A two-drive version (26-6002) with a total 2½-Megabyte storage capacity will be available for \$5798*.

The new TRS-80 Model 16 uses two microprocessors: the Z-80A (used in the TRS-80 Model I, Model II and Model III microcomputers); and a state-of-the-art 16-bit microprocessor, the MC68000, developed by Motorola.

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The MC68000 microcomputer offers several advantages. First, it allows the Model 16 to access much more memory than systems using 8-bit microprocessors. The Model 16 comes with 128K of RAM, but can be expanded in 128K steps to 512K (½-Megabyte). Second, because the data path is a full sixteen bits wide, and because the MC68000 can perform internal 32-bit operations, more complex data can be processed, and at much higher speeds. Together, these advantages give the Model 16 the capability to process more sophisticated programs. This means that the compact, self-contained Model 16 can do many jobs that have been too complex for ordinary microcomputers.

Using the Z-80A as a second microprocessor brings additional advantages, including software compatibility with existing TRS-80 Model II software. Operationally, the Z-80A relieves the MC68000 of time-consuming "housekeeping chores" (like input/output control, for example), freeing the main CPU to use its superior processing power to its fullest.

Model II software compatibility means the Model 16 can put Radio Shack's substantial library of off-the-shelf Model II software to work immediately. Model 16 COBOL will be available at the same time as the Model 16; new software that will be able to take full advantage of the Model 16's advanced capabilities (including FORTRAN and both compiler and interpreter BASIC) will be announced shortly.

The TRS-80 Model 16 has been designed to permit its expansion to multi-user operation with the addition of one or two low-cost terminals, which may be located remotely. This lets as many as three users access programs and information at the same time, with no perceptible loss of performance.

Low-cost video data terminals (like Radio Shack's just-announced DT-1), can

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be used as remote workstations for the Model 16, as can TRS-80 Model II computers. This demonstrates a clear route to cost-effective system upgrading for installations currently using Model II computers.

The Model 16 also offers TRS-80 hardware compatibility. It's equipped with two RS-232C serial interfaces ports and a parallel interface port. One of its serial ports is capable of bisynchronous communications to IBM and other mainframes. A hard disk port may be added to permit the use of up to four of Radio Shack's 8.4-Megabyte hard disk drives.

A Model 16 system has the option of growing with a wide variety of "off-the-shelf" TRS-80 peripherals, including printers, plotters, digitizers, telephone modems and more.

The TRS-80 Model 16 is compatible with Radio Shack's previously-announced ARCNET local network. TRS-80 ARCNET will support any combination of as many as 255 Model II and Model 16 computers.

The TRS-80 Model 16 comes with thorough and easy to understand manuals and the Radio Shack TRSDOS operating system, which includes an assembly language development system.

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*Prices mentioned above are available at Radio Shack stores, Computer Centers and participating dealers.